LLM Router - Benchmark Evaluation Report

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1. EXECUTIVE SUMMARY & VERDICT

Verdict: Router reduced mean latency by 24.3% vs heavy-only (KPI-2 target of 30% lower not met).

Hardware: linux x64; AMD Ryzen 7 7435HS; 24.9 GB RAM; GPU: NVIDIA GeForce RTX 4050 Laptop GPU (6GB VRAM)

Total Prompts: 72 labeled test prompts

Date Executed: 10/23/2026, 11:30:36 PM

2. KEY PERFORMANCE INDICATORS (KPIS)

â¢ KPI-1 Routing Accuracy: 100% (Target: 85% | Pass: YES)

â¢ KPI-2 Latency Reduction: 24.3% (Target: 30% | Pass: NO)

â¢ KPI-3 Cache Hit Rate (Run C): 58%

3. DETAILED RUN METRICS SUMMARY

Metric	Run A (Heavy Only)	Run B (Router)	Run C (Router+Cache
Mean Latency	29.9 s	22.6 s	8.61 s
p50 Latency	31.3 s	21.4 s	46 ms
p95 Latency	66.3 s	56.9 s	44.1 s
TTFT Mean	227 ms	751 ms	143 ms
Tokens / Sec	15.0 tok/s	58.4 tok/s	60.8 tok/s
Model Swaps	72	72	60
Cache Hit Rate	0%	0%	58%
Routing Accuracy	-	100%	100%
Total Wall Time	2151.4 s	1628.5 s	1300.2 s

4. ROUTING CONFUSION MATRIX (Run B)

Expected \ Routed	light	code	heavy
light	24	0	0
code	0	24	0
heavy	0	0	24

5. METHODOLOGY & PROVENANCE

Backend: Ollama local inference. Order: deterministic prompt-file order; A = forced heavy/no-cache, B = auto/no-cache, C = auto/cache-on over two full passes. Warm-up: before each run, one unmeasured forced-heavy request is completed; identical prompt and sampling settings are used. Cache is flushed before each run; for C it remains enabled for both passes. Cache results include labeled duplicate prompts. Latency is measured at the gateway from the successful request attempt through terminal SSE event; rate-limit backoff is excluded from per-request latency but included in total wall time. Router TTFT/tokens-per-second are reported from final usage metadata. Swap count is a proxy for non-cache-hit inference requests with positive final-chunk load_ms; it

is not an independent hardware swap counter. Routing accuracy and confusion matrix use the dry-run preview endpoint and expected tiers in prompts.json; this is not answer-quality evaluation. GPU/VRAM values are omitted when the stats contract reports them as unavailable. Mock-backend measurements are for smoke testing only and are not hardware performance claims.